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CAMEF: Causal-Augmented Multi-Modality Financial Forecasting

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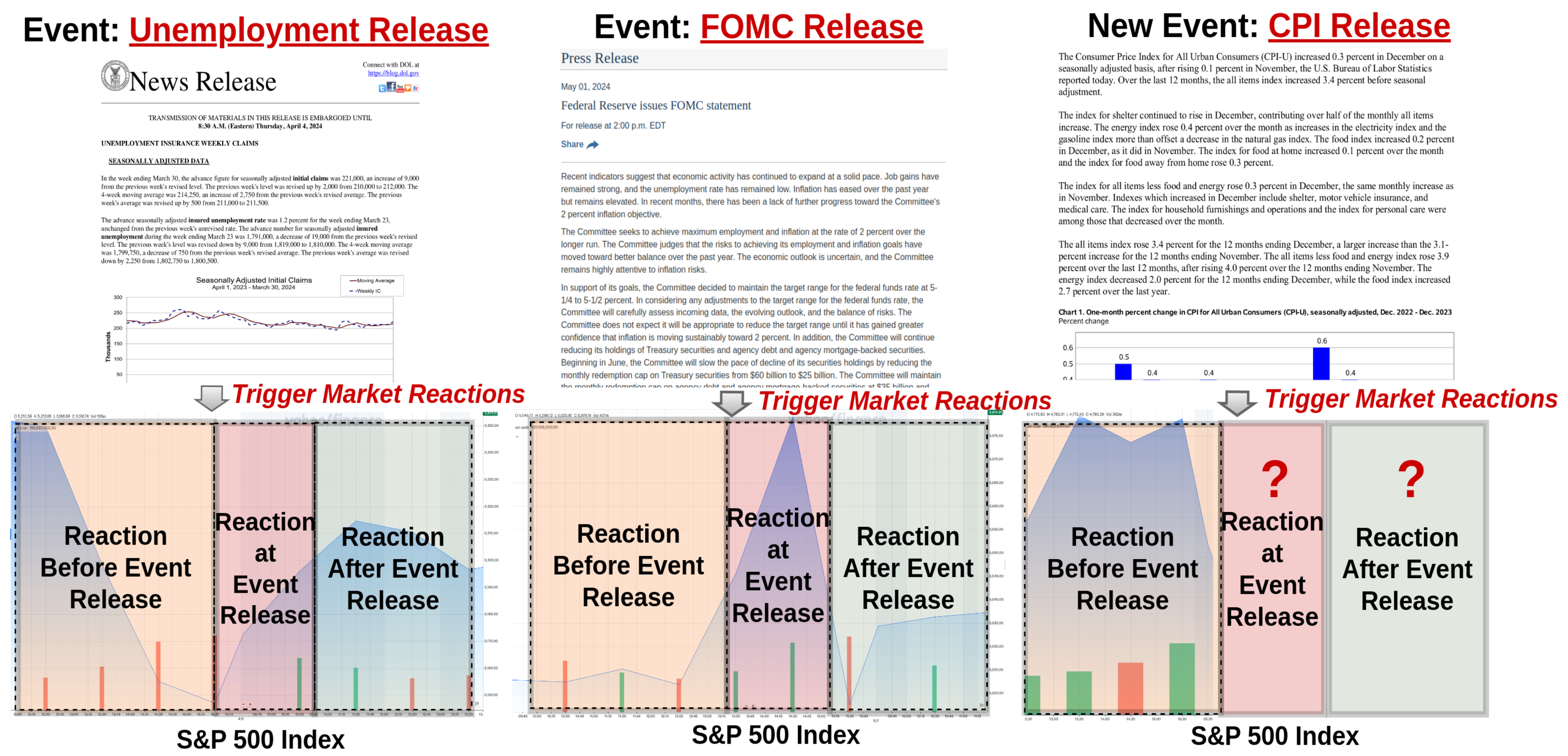
Motivation

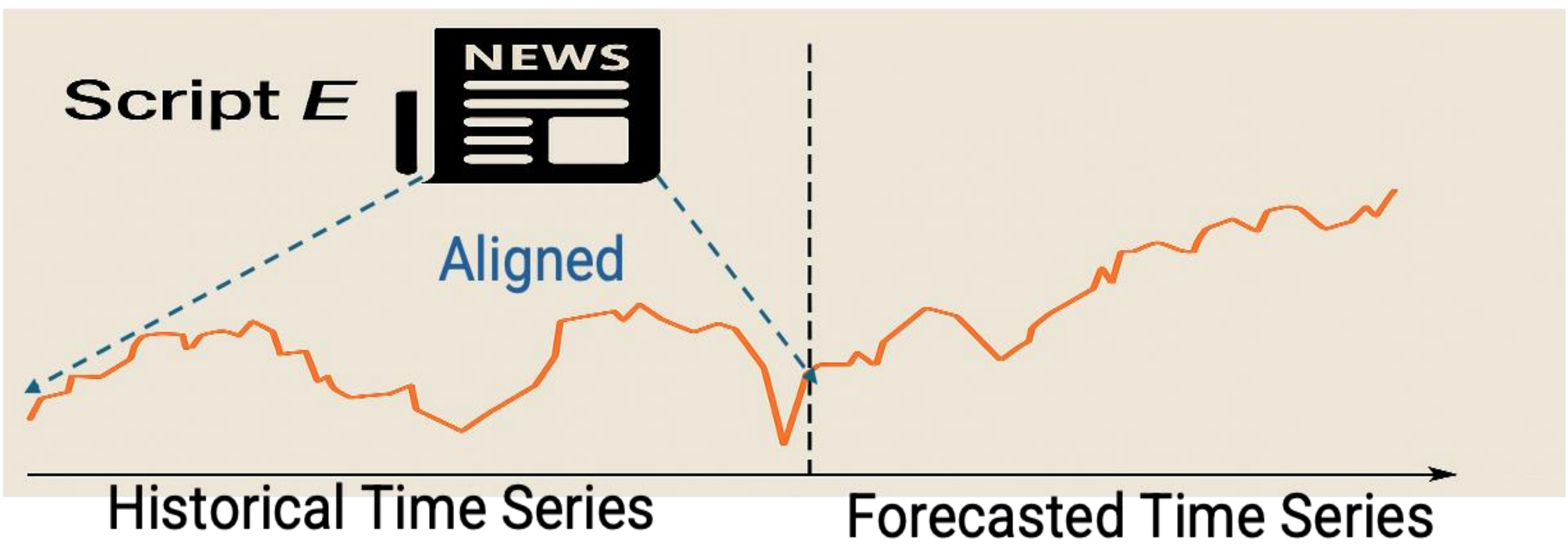
- ➢ **Salient macroeconomic events often trigger major market reactions**
- ➢ **We aim to predict market movement from upcoming key events**



(a) SP500 Index Reaction to Unemployment Release
(b) SP500 Index Reaction to FOMC Meeting
(c) Prediting the SP500 Index Reaction to new event

Problem Definition

Given a dataset U = $\{[X_{i-\tau:i+\tau} \mapsto E_i]\}^{n}_{i=1}$ consisting of n aligned event and time-series pairs, for each data pair in U, the model uses both the event text E_i and the historical time-series segment $X_{i-\tau:i}$, which spans τ steps before the event's release at time i, to forecast the future time steps $X_{i+1:i+\tau}$.



Data Collection

- ❑ **Textual Modality Data**

6 event types (FOMC, CPI, etc.) from 1993–2024; 2,000+ event scripts and 25,000+ counterfactuals (via LLaMA3)

Event Type	Data Type	Frequency	Period	No. of Events	No. of C.F.s	Source
FOMC	Html	Quarterly	1993.3 ~2024.6	255	2,550	www.federalreserve.gov
Unemployment Insurance Claims	PDF, Txt	Weekly	2002.10 ~2024.6	913	9,130	oui.doleta.gov
Employment Situation	Html, Txt	Monthly	1994.2 ~2024.6	363	3,630	www.bls.gov
GDP Advance Report	Html	Monthly	1996.8 ~2024.6	333	3,330	www.bea.gov
CPI Report	Html, Txt	Monthly	1994.2 ~2024.6	357	3,570	www.bls.gov
PPI Report	Html, Txt	Monthly	1994.2 ~2024.6	348	3,480	www.bls.gov

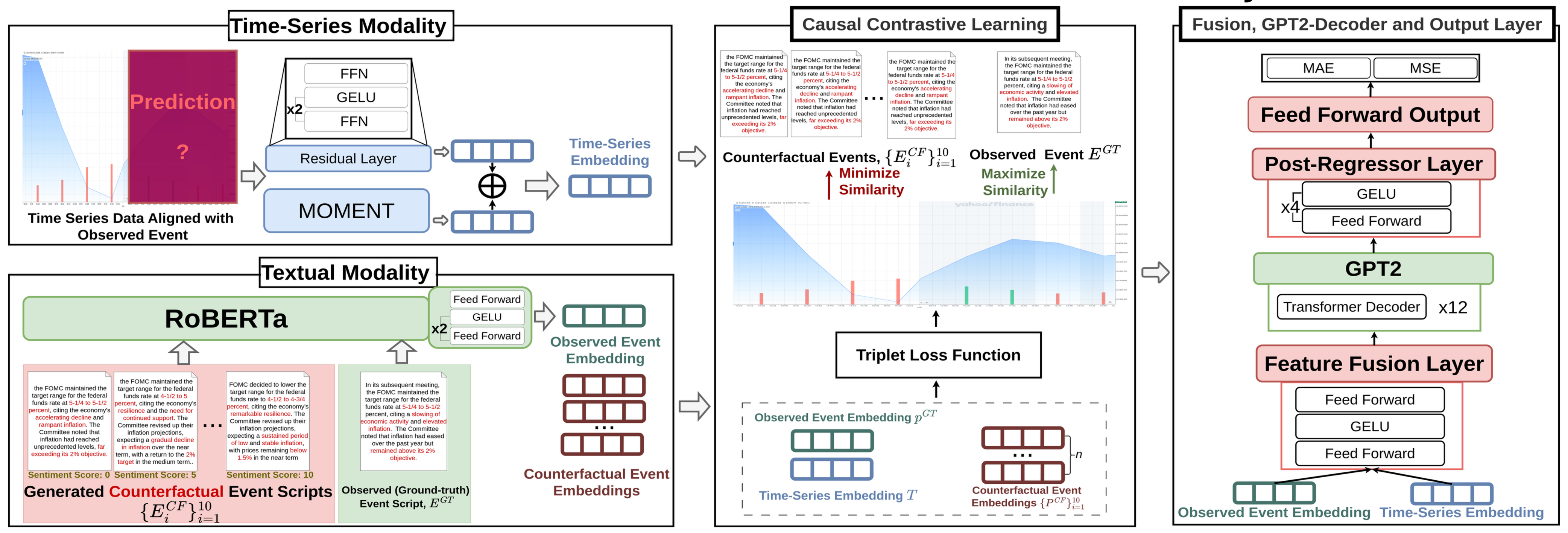
- ❑ **Time-Series Modality Data:**

5 high frequency trading data major US stock indices and bonds; >2.4M points from 2008–2024

Time Series Data	Data Types	Frequency	Range	No. of Data Points
SP500 (SPX)	Open, Close, High, Low	5 Min	2008.01 ~2024.06	331,257
Dow Industrial (INDU)	Open, Close, High, Low	5 Min	2012.07 ~2024.06	263,445
NASDAQ (NDX)	Open, Close, High, Low	5 Min	2008.01 ~2024.06	332,616
US Treasury Bond at 1-Month (USGG1M)	Open, Close, High, Low	5 Min	2013.01 ~2024.06	751,443
US Treasury Bond at 5-Year (USGG5YR)	Open, Close, High, Low	5 Min	2013.01 ~2024.06	734,773

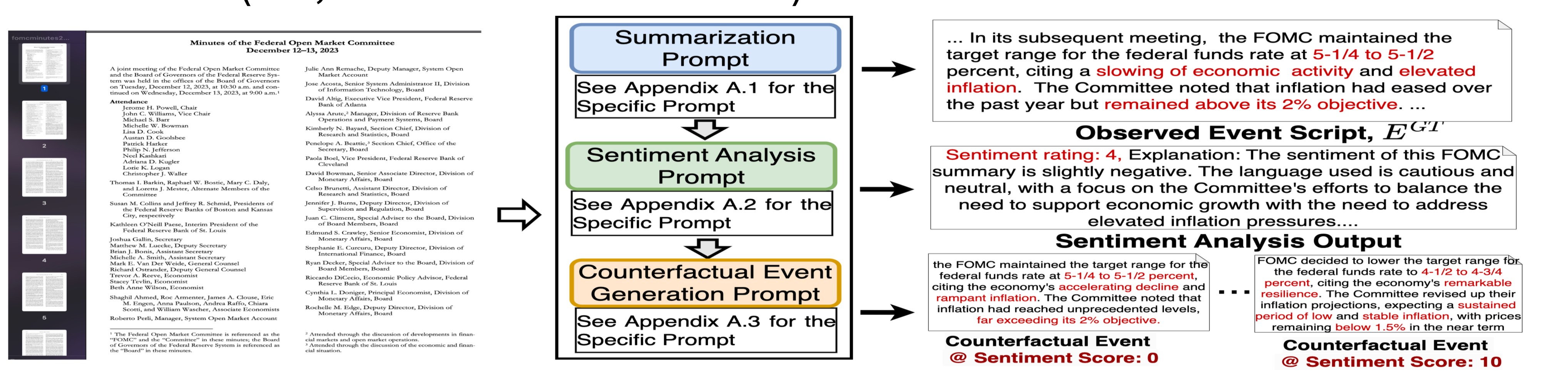
Model Framework

- ❑ **Encoders:** RoBERTa (text), MOMENT (time series)
- ❑ **Causal Learning:** Helps learning **causal vs. counterfactuals**
- ❑ **Fusion:** MLP
- ❑ **Decoder:** GPT2-style



Counterfactual Generation

- ❑ Propose structured prompt strategies to generate sentiment-shifted variations (i.e., counterfactual events)



Original FOMC Meeting Report
Prompt Pipeline via LLaMa3
Generated Output from LLM

Experiment

- ● **Overall Performance:**

CAMEF outperforms both SOTA unimodal and multimodal baselines

Model / Datasets	Forecasting Length	SP500 (SPX) MSE	SP500 (SPX) MAE	Dow Industrial (INDU) MSE	Dow Industrial (INDU) MAE	NASDAQ (NDX) MSE	NASDAQ (NDX) MAE	USGG1M MSE	USGG1M MAE	USGG5YR MSE	USGG5YR MAE
ARIMA	35	0.0032628	0.0308016	0.0121513	0.0523810	0.0065253	0.0435694	0.0072223	0.0262512	0.0038973	0.0340227
	70	0.0035361	0.0352004	0.0139245	0.0656002	0.0082324	0.0520933	0.0028710	0.0304132	0.0039441	0.0366630
	140	0.0051080	0.0439935	0.0219147	0.0793471	0.0118692	0.0665155	0.0089949	0.0338275	0.0050746	0.0455617
DLinear	35	0.0144331	0.0896910	0.0395136	0.1380539	0.0183989	0.0999178	0.0108576	0.0706170	0.0147211	0.0876379
	70	0.0120578	0.0817373	0.0406573	0.1282699	0.0189431	0.0972439	0.0093591	0.0678380	0.0146430	0.0881574
	140	0.0178138	0.0931039	0.0747210	0.1724027	0.0344153	0.1287803	0.0101465	0.0645396	0.0179185	0.0977673
Autoformer	35	0.0068136	0.0540556	0.0277636	0.0975948	0.0135249	0.0796486	0.0047505	0.0388076	0.0092717	0.0622862
	70	0.0088997	0.0628341	0.0375279	0.1185710	0.0185264	0.0933398	0.0065554	0.0471531	0.0113750	0.0727373
	140	0.0158248	0.0829188	0.0772580	0.1640365	0.0334504	0.1262163	0.0086212	0.0533337	0.0152298	0.0875821
FEDformer	35	0.0072377	0.0576221	0.0304808	0.1044447	0.0128668	0.0758742	0.0063596	0.0519141	0.0094995	0.0494306
	70	0.0088056	0.0621386	0.0399824	0.1229772	0.0171008	0.0889995	0.0062841	0.0448822	0.0099615	0.0664197
	140	0.0157429	0.0819469	0.0786426	0.1677705	0.0313433	0.1214097	0.0083784	0.0518365	0.0131231	0.0782861
iTransformer	35	0.0064209	0.0516341	0.0270860	0.0927605	0.0125008	0.0751656	0.0011000	0.0155975	0.0056660	**0.0183524**
	70	0.0069612	0.0540920	0.0304566	0.1038424	0.0151214	0.0816262	0.0021811	0.0221315	**0.0011721**	**0.0226975**
	140	0.0128021	0.0718771	0.0680991	0.1486782	0.0254479	0.1047119	0.0052255	0.0327429	**0.0017441**	**0.0282537**
PatchTST	35	0.0063304	0.0507462	0.0293131	0.0989455	0.0122679	0.0764552	0.0012060	0.0163610	0.0063078	0.0520734
	70	0.0072471	0.0547738	0.0339444	0.1116023	0.0153753	0.0824677	0.0021643	0.0223036	0.0079617	0.0606007
	140	0.0130219	0.0712171	0.0688582	0.1452592	0.0256001	0.1047749	0.0054544	0.0341517	0.0118553	0.0747537
GPT4MTS	35	0.00795088	0.0674255	0.0026558	0.0417553	0.0011035	0.0240469	0.0017016	0.0309320	0.0028713	0.0389277
	70	0.00171038	0.0305950	0.0027033	0.0393112	0.0016205	0.0336116	0.0019098	0.0279222	0.0023371	0.0354777
	140	0.00212612	0.0330497	0.0045029	0.0458267	0.0025175	0.0341671	0.0013648	0.0260887	0.0037004	0.0449272
TEST	35	0.00073333	0.0199733	0.0026572	**0.0296887**	0.0006593	0.0194091	0.0003252	0.0137511	0.0013633	0.0265036
	70	0.00078762	0.0205412	0.0088903	0.0599179	0.0010678	0.0248594	0.0010515	0.0182706	0.0034630	0.0415002
	140	0.00278572	0.0467150	0.0070749	0.0557744	0.0020130	0.0362325	0.0006995	0.0207850	0.0024356	0.0375164
CAMEF	35	**0.00048860**	**0.0154050**	**0.0025349**	0.0366245	**0.0005468**	**0.0178845**	**0.0002883**	**0.0118010**	**0.0013234**	0.0260618
	70	**0.00064780**	**0.0178691**	**0.0025042**	**0.0365500**	**0.0005814**	**0.0162882**	**0.0004402**	**0.0139699**	0.0020701	0.0326371
	140	**0.0010756**	**0.0210284**	**0.0039313**	**0.0383459**	**0.0010159**	**0.0207716**	**0.0004938**	**0.0148485**	0.0022458	0.0336680

- ● **Ablation Study:**

All components (fusion, loss, decoder) contribute

Textual	Causal	Feature Fusion	GPT2 Decoder	Post-Regressor	SPX 35	SPX 70	SPX 140	INDU 35	INDU 70	INDU 140	NDX 35	NDX 70	NDX 140	USGG1M 35	USGG1M 70	USGG1M 140
✗	✓	✓	✓	✓	0.00074	0.00340	0.00120	0.00340	0.00330	0.01033	0.00131	0.00097	0.00197	0.00080	0.00179	0.00092
✓	✗	✓	✓	✓	0.00068	0.00095	0.00106	0.02939	0.00281	0.00533	0.00064	0.00073	0.00121	0.00065	0.00083	0.00099
✓	✓	✗	✓	✓	0.00080	0.00079	0.00110	0.01173	0.00391	0.00581	0.00069	0.00073	0.00168	0.00047	0.00046	0.00216
✓	✓	✓	✗	✓	0.00073	0.00067	0.00114	0.26768	0.72387	0.18091	0.00062	0.00069	0.00158	0.00043	0.02110	0.75057
✓	✓	✓	✓	✗	0.00662	0.03911	0.00979	0.50841	0.57007	0.22267	0.00774	0.00852	0.00950	0.28932	0.02110	0.75057
Full CAMEF Model					**0.00048**	**0.00064**	**0.00107**	**0.00253**	**0.00250**	**0.00393**	**0.00054**	**0.00058**	**0.00101**	**0.00028**	**0.00044**	**0.00049**

- ● **Event Importance Test:**

FOMC is most influential among all event types

Event Type	Forecasting Length = 35 MSE	Forecasting Length = 35 MAE	Forecasting Length = 70 MSE	Forecasting Length = 70 MAE	Forecasting Length = 140 MSE	Forecasting Length = 140 MAE
Unemployment Insurance	0.0004870 ↓	0.0159497 ↑	0.0005199 ↓	0.0157778 ↓	0.0006948 ↓	0.0190141 ↓
Employment Situation	0.0003923 ↓	0.0142684 ↓	0.0004612 ↓	**0.0154431** ↓	0.0013767 ↑	0.0218677 ↑
GDP Adcance	0.0006203 ↑	0.0175397 ↑	0.0005897 ↓	0.0175464 ↓	0.0011554 ↑	0.0217548 ↑
FOMC Minutes	**0.0003401** ↓	**0.0127694** ↓	**0.0004448** ↓	0.0170094 ↓	**0.0006433** ↓	**0.0185657** ↓
CPI Report	0.0005645 ↑	0.0160723 ↑	0.0010660 ↑	0.0212536 ↑	0.0008187 ↓	0.0197824 ↓
PPI Report	0.0005275 ↑	0.0148434 ↓	0.0008054 ↑	0.0201844 ↑	0.0017646 ↑	0.0251856 ↑
Full Selection	0.0004886	0.0152405	0.0006478	0.0178691	0.0010756	0.0210284

Conclusion

- ➢ First framework to align policy text and market data causally
- ➢ Introduced multimodal financial event dataset with LLM-generated counterfactuals
- ➢ Achieved state-of-the-art results with interpretability

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